

Real Algebra

Real Closed Fields and Galois-Theoretic Characterization

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Algebraic Background of Ordered Fields (Artin–Schreier)

Orderings and Preorderings of Fields

Definition (Ordering of a field)

Let K be a field. An **ordering** of K is a subset $P \subseteq K$ satisfying:

01 $P + P \subseteq P$ and $P \cdot P \subseteq P$

02 $P \cap (-P) = \{0\}$

03 $P \cup (-P) = K$

The pair (K, P) is an **ordered field**.

Notation: $P + P := \{a + b : a, b \in P\}$ and $P \cdot P := \{ab : a, b \in P\}$.

Remark on (O2)

If P is closed under addition and multiplication and satisfies (O3), then the condition (O2) is equivalent to

$$(O2') \quad -1 \notin P.$$

Algebraic Background of Ordered Fields

Orderings and Preorderings of Fields

Definition (Preordering of a Field)

A **preordering** of a field K is a subset $T \subseteq K$ satisfying:

- (P1) $T + T \subseteq T$ and $T \cdot T \subseteq T$
- (P2) $T \cap (-T) = \{0\}$
- (P3) $\forall a \in K, a^2 \in T$

Examples

- The set of non-negative reals $\mathbb{R}_{\geq 0}$ is the unique ordering of $\mathbb{R}$.
- Similarly, $\mathbb{Q}_{\geq 0}$ is the unique ordering of $\mathbb{Q}$.
- $\mathbb{C}$ *cannot* be ordered because -1 can be expressed as a sum of squares ($-1 = i^2$).

Algebraic Background of Ordered Fields

Properties of Preorderings & Sums of Squares

Remarks on Preorderings

- Similarly, assuming (P1) and (P3), condition (P2) is equivalent to (P2') $-1 \notin T$.
- Every ordering is a preordering. Indeed, (P3) is a weaker version of (O3), where $\forall a \in K, (a \in P \vee -a \in P)$ implies $a^2 \in P$ by (O1).
- Arbitrary intersections of preorderings are again preorderings.
- Let $\mathcal{T} = (T_\alpha : \alpha \in I)$ be a nonempty family of preorderings of a field K .
 - The intersection $\bigcap_{\alpha \in I} T_\alpha$ is also a preordering.
 - If $\mathcal{T}$ is **upward directed** (i.e., for all $\alpha, \beta \in I$ there exists $\gamma \in I$ such that $T_\alpha \cup T_\beta \subseteq T_\gamma$), then the union $\bigcup_{\alpha \in I} T_\alpha$ is also a preordering.
 - In particular, if K has a preordering, then it will have a smallest one.

Ref: M. Knebusch & C. Scheiderer, *Real Algebra: A First Course* (2022)



Algebraic Background of Ordered Fields

The Smallest Preordering

The Smallest Preordering

We denote the set of sums of squares in K by $\sum K^2$:

$$\sum K^2 := \{a_1^2 + \cdots + a_n^2 \mid n \in \mathbb{N}, a_1, \dots, a_n \in K\}$$

$\sum K^2$ is contained in every preordering of K . And It is a preordering if and only if $-1 \notin \sum K^2$.

Ref: M. Knebusch & C. Scheiderer, Real Algebra. A First Course (2022)

Algebraic Background of Ordered Fields

Real Fields

Definition (Real Field)

The field K is called **real** if $-1 \notin \sum K^2$, i.e., when -1 cannot be written as a sum of squares in K .

Immediate Consequences

- A field K has a preordering **if and only if** it is real.
- In this case, $\sum K^2$ is the *smallest* preordering on K .
- All real fields must have characteristic 0.

Ref: M. Knebusch & C. Scheiderer, *Real Algebra. A First Course* (2022)

Algebraic Background of Ordered Fields

Lemma 1.1.8 - Enlarging a Preordering

Lemma (Lemma 1.1.8)

Let T be a preordering of K and let $a \in K$ with $a \notin T$. Then

$$T - aT = \{b - ac \mid b, c \in T\}$$

is again a preordering of K .

Proof Sketch.

$T - aT$ clearly satisfies (P1) and (P3). We only need to show $-1 \notin T - aT$:

- Assume that $-1 \in T - aT$, so $-1 = b - ac$.
- This implies $c \neq 0$ (otherwise $-1 = b \in T$, impossible).
- Rearranging gives $ac = 1 + b$, so $a = c^{-2} \cdot c(1 + b)$.
- Since $c^{-2}, c, 1, b \in T$, we get $a \in T$, contradicting $a \notin T$. □

Algebraic Background of Ordered Fields

Lemma 1.1.9 - Existence of Orderings

Lemma (Lemma 1.1.9)

For every preordering T of K there exists an ordering P of K such that $T \subseteq P$.

Proof.

- Let $M := \{T' \subseteq K \mid T' \text{ is a preordering and } T \subseteq T'\}$.
- M is ordered by inclusion, $M \neq \emptyset$, and Zorn's Lemma applies.
- Thus, there exists a **maximal preordering** P of K with $T \subseteq P$.
- **Why is P an ordering?** If $a \in K$ and $a \notin P$, then by Lemma 1.1.8, $P - aP$ is a preordering containing P .
- By the maximality of P , we must have $P = P - aP$.
- Since $-a \in P - aP$, it follows that $-a \in P$. Thus $P \cup (-P) = K$, meaning P is a full ordering. □

Definition (Real Closed Field)

A field K is called **real closed** if it is real and has no proper algebraic extension that is real.

- **Maximality:** The field K has no proper algebraic extension that is real. If we expand K algebraically to any strictly larger field $L \supsetneq K$, the new field L will *always* break the real property (i.e., -1 becomes a sum of squares in L $-1 \in \sum L^2$).
And K is at the absolute limit of “realness.”

Example

The field $\mathbb{R}$ of real numbers is a well-known example of a real closed field.

Ref: M. Knebusch & C. Scheiderer, Real Algebra. A First Course (2022)

Characterization of Real Closed Fields

Proposition (proposition 1.5.2 - Characterization of Real Closed Fields)

Let K be a field. The following statements are equivalent:

- (i) K is real closed;*
- (ii) there exists an ordering P of K that cannot be extended to any proper algebraic extension of K ;*
- (iii) $K^{*2} \cup \{0\} = \{a^2 : a \in K\}$ is an ordering of K , and every polynomial (in one variable) of odd degree over K has a zero in K .*

Uniqueness of Ordering

Moreover, if these conditions are satisfied, then $K^{*2} \cup \{0\}$ is the **only** ordering of K .

Proof: The final comment is clear by (iii) (every ordering P satisfies $K^{*2} \cup \{0\} \subseteq P$).

Background: Extension of Orderings

Before we prove the characterization theorem, we must recall what it means to extend an ordering, which will be crucial for the upcoming proofs.

Definition (Definition 1.3.1)

Let L/K be a field extension. An ordering P of K **extends** to an ordering Q of L if

$$P \subseteq Q$$

(and consequently, $P = K \cap Q$).

Ref: M. Knebusch & C. Scheiderer, Real Algebra. A First Course (2022)

Background: Extension of Orderings

Proposition 1.3.2 - Equivalence

Proposition (Proposition 1.3.2)

Let L/K be a field extension and P an ordering of K . The following statements are equivalent:

- i) P extends to an ordering Q of L ;*
- ii) -1 is not contained in the semiring T generated by P and $\sum L^2$ in L (i.e., the smallest subset T of L such that $P \cup \sum L^2 \subseteq T$, $T + T \subseteq T$ and $TT \subseteq T$);*
- iii) every quadratic form $\langle p_1, \dots, p_n \rangle$ with $p_1, \dots, p_n \in P^* := P \setminus \{0\}$ is anisotropic over L .*

Background: Extension of Orderings

Proposition 1.3.2 - Proof Sketches

Proof Highlights

- (i) $\implies$ (iii): If $p_1 b_1^2 + \cdots + p_n b_n^2 = 0$ with $b_i \in L$, then $b_1 = \cdots = b_n = 0$ because Q is an ordering (sum of positive elements is zero only if all are zero).
- (iii) $\implies$ (ii): Elements of T take the form $p_1 b_1^2 + \cdots + p_m b_m^2$. If $-1 \in T$, we can manipulate it to find an isotropic form over L , contradicting (iii).
- (ii) $\implies$ (i): By assumption, T is a preordering of L . By earlier results (Lemma 1.1.9), every preordering is contained in an ordering Q of L , which extends P .

Background: Extension of Orderings

Quadratic Extensions

Proposition (Proposition 1.3.3)

Let P be an ordering of K , $d \in K$ and $L = K(\sqrt{d})$. Then P extends to L if and only if $d \in P$.

Proof Intuition

- ($\implies$) If P extends to an Ordering Q of L , then $\sqrt{d} \in L$ must be positive in the extended ordering Q . Hence, $d = (\sqrt{d})^2 \in Q$, and since $P \subseteq Q$, we have $d \in P$.
- ($\impliedby$) If $d \in P$, assume for contradiction that P does *not* extend.
 - By Proposition 1.3.2, this forces -1 to be a sum of squares over L :
$$-1 = \sum p_i (a_i + b_i \sqrt{d})^2$$
 - Expanding this and matching the coefficients with respect to the K -basis $\{1, \sqrt{d}\}$ of L gives: $-1 = \sum p_i (a_i^2 + db_i^2)$
 - Since p_i, d , and all squares are in P , this equation forces $-1 \in P$. This is a contradiction, so P must extend!

Background: Extension of Orderings

Odd Degree Extensions

Proposition (Proposition 1.3.5)

Let L/K be a finite field extension of odd degree. Then *every* ordering of K extends to L .

Symbolic Proof Sketch (via Prop 1.3.2)

Assume there exist an ordering P of K that does *not* extend to L .

- **Prop 1.3.2(ii)** $\implies \exists p_i \in P, x_i \in L$ such that: $-1 = \sum p_i x_i^2$
- Since L/K is odd, that implies the form is already isotropic over the base field K :

$$\exists y_i \in K : y_0^2 + \sum p_i y_i^2 = 0$$

- $q_j \in P$ and $y_j^2 \in P \implies \sum q_j y_j^2 \in P$
- $\implies -1 \in P$ (**Contradiction!**)

Characterization of Real Closed Fields

Proof of Proposition 1.5.2: (i) $\implies$ (ii) and (ii) $\implies$ (iii)

Proof of (i) $\implies$ (ii).

This is clear since a field has an ordering if (and only if) it is real. $\square$

Proof of (ii) $\implies$ (iii).

- Assume for contradiction that $K^{*2} \cup \{0\} \neq P$.
- Then there would exist some $d \in P$ with $\sqrt{d} \notin K$.
- By **Proposition 1.3.3**, the ordering P would extend to the proper algebraic extension $K(\sqrt{d})$. This contradicts assumption (ii)!
- Furthermore, K has no proper extensions of odd degree by **Proposition 1.3.5**. $\square$

Characterization of Real Closed Fields

Proof of Proposition 1.5.2: (iii) $\implies$ (i) [Part 1]

Proof of (iii) $\implies$ (i).

Let $L \supset K$ be a finite proper field extension. We must show L is not real.

- By (iii), K has no odd degree extensions, so $[L : K]$ is a power of 2.
- Since K has an ordering, $\text{char } K = 0$, meaning L/K is separable.
- Let L_1 be the Galois closure of L over K . Let $G = \text{Gal}(L_1/K)$ and $H = \text{Gal}(L_1/L)$.
- By Galois theory, $\bigcap_{g \in G} H^g = \{1\}$, where $H^g := g^{-1}Hg$.
- By elementary group theory, $|G|$ divides $\prod_{g \in G} [G : H^g]$.
- Since these indices are powers of 2, G is a 2-group.



Characterization of Real Closed Fields

Proof of Proposition 1.5.2: (iii) $\implies$ (i) [Part 2]

Proof continued...

- By Sylow's Theorems, there exists a subgroup $\tilde{H}$ of G of index 2 with $H \subset \tilde{H}$.
- Let F be the fixed field of $\tilde{H}$. Then $K \subset F \subset L$ and $[F : K] = 2$.
- Because the degree is 2, $F = K(\sqrt{a})$ for some $a \in K$.
- Since $K^{*2} \cup \{0\}$ is an ordering (by assumption iii), and $a \notin K^{*2} \cup \{0\}$, it must be negative. Thus $a = -b^2$ for some $b \in K^*$.
- Therefore in F : $-1 = a/b^2 = (\sqrt{a}/b)^2$.
- Since -1 is a square in F , F is not real. We conclude that L (which contains F) is not real either. □

The Fundamental Theorem of Algebra

Theorem 1.5.4

Theorem (Fundamental Theorem of Algebra)

If R is a real closed field, then $R(\sqrt{-1})$ is algebraically closed.

Proof (Gauss) - Part 1.

Let $C := R(i)$ where $i := \sqrt{-1}$. We use Proposition 1.5.2(iii).

- Every finite extension of R (and so also every finite extension of C) has 2-power degree.
- Assume for contradiction that C is not algebraically closed.
- As in the proof of Prop 1.5.2, C has an extension F with $[F : C] = 2$.
- Thus, there exist $a, b \in R$ such that $\alpha := a + bi$ is not a square in C . Hence, $b \neq 0$.



The Fundamental Theorem of Algebra

Proof (Gauss) - Part 2

Proof continued...

- Since R^2 is an ordering of R , there exists $c \in R$ with $a^2 + b^2 = c^2$ and $c > 0$.
- We can find $x, y \in R$ with $x > 0$, $\text{sign } y = \text{sign } b$, and:

$$x^2 = \frac{c+a}{2}, \quad y^2 = \frac{c-a}{2}$$

- Now evaluate $(x + iy)^2 = (x^2 - y^2) + 2xyi$.
- Notice $x^2 - y^2 = a$, and $(2xy)^2 = 4x^2y^2 = c^2 - a^2 = b^2$.
- Since $\text{sign}(xy) = \text{sign } b$, we have $2xy = b$.
- It follows that $(x + iy)^2 = a + bi = \alpha$, which contradicts that α is not a square in C . □

Relative Algebraic Closure

Corollary (Corollary 1.5.5)

*Let R be a real closed field and let $K \subseteq R$ be a subfield. Then K is real closed **if and only if** K is (relatively) algebraically closed in R .*

Proof.

Assume K is algebraically closed in R . Let $i = \sqrt{-1}$. Then $K(i)$ is algebraically closed in $R(i)$.

- If $\alpha = a + bi \in R(i)$ is algebraic over $K(i)$ (where $a, b \in R$), it is algebraic over K .
- The conjugate $\bar{\alpha} = a - bi$ is also algebraic.
- Thus $a = (\alpha + \bar{\alpha})/2$ and $b = i(\bar{\alpha} - \alpha)/2$ are algebraic over K .
- Since K is algebraically closed in R , $a, b \in K$, hence $\alpha \in K(i)$.
- By Theorem 1.5.4, $K(i)$ is algebraically closed, meaning K is real closed.

Rigidity of Real Closed Fields

Another characteristic of real closed fields is their relative rigidity:

Proposition (Proposition 1.5.6)

Let R be a real closed field.

- Ⓐ *Every (ring) endomorphism φ of R is **order preserving**, i.e., $a \leq b \implies \varphi(a) \leq \varphi(b)$ for all $a, b \in R$.*
- Ⓑ *If $K \subseteq R$ is a subfield and R/K is algebraic, then the identity is the **only** K -automorphism of R .*

Notation Convention

If R is real closed, we will henceforth denote the uniquely determined ordering of R by $\leq$ without further comment.

Rigidity of Real Closed Fields

Proof of Proposition 1.5.6

Proof of (a).

This follows immediately from the fact that the unique ordering consists exactly of the squares: $R^2 = \{a \in R \mid a \geq 0\}$.

(Note: Endomorphisms preserve squares, so if $b - a = c^2 \geq 0$, then $\varphi(b) - \varphi(a) = \varphi(c^2) = (\varphi(c))^2 \geq 0$, preserving the order.) □

Proof of (b).

Let $\varphi \in \text{Aut}_K(R)$ and $a \in R$.

- Since R/K is algebraic, the set of iterates $\{\varphi^n(a) \mid n \in \mathbb{N}\}$ must be a finite set.
- Assume for contradiction that $\varphi(a) \neq a$. Without loss of generality, let $\varphi(a) > a$.
- Since φ is order preserving by (a), applying it repeatedly yields:

$$a < \varphi(a) < \varphi^2(a) < \dots$$

Galois Theoretic Characterization of Real Closed Fields

Artin-Schreier Theorem (1927)

Theorem (1.6.1 - E. Artin & O. Schreier)

Let K be a field with algebraic closure $\overline{K}$.

If $K \neq \overline{K}$ and $[\overline{K} : K] < \infty$, then K is real closed and $\overline{K} = K(\sqrt{-1})$.

Proof Strategy (J. Leicht)

Let $n = [\overline{K} : K]$. The proof is broken down into 6 structural steps:

- 1 Show K is perfect (forces $\overline{K}/K$ to be finite Galois).
- 2 Analyze a maximal intermediate field L with prime index $q = [\overline{K} : L]$ and show $\text{char } K \neq q$.
- 3 Prove $q = 2$.
- 4 Deduce $\overline{K} = L(\sqrt{-1})$.
- 5 Show $L = K$, forcing $\overline{K} = K(\sqrt{-1})$.
- 6 Conclude K is real (and therefore real closed).

Galois Theoretic Characterization of Real Closed Fields

Proof of Theorem 1.6.1: Steps 1 & 2

(1) K is perfect

- Assume not $\implies \text{char } K = p > 0, \exists a \in K$ with $\alpha := a^{1/p} \notin K$.
- Iterative norm arguments $N_{K(\alpha)/K}$ yield infinite field extensions of degrees $p, p^2, p^3, \dots$
- **Contradiction:** $[\bar{K} : K] < \infty$.
- $\therefore K$ is perfect $\implies \bar{K}/K$ is a finite Galois extension.

(2) $\text{char } K \neq q$

Let L be a maximal intermediate field, and $q := [\bar{K} : L]$ (which is prime).

- Assume $\text{char } K = p = q$.
- Artin-Schreier theory on map $\psi(b) := b^p - b$ and trace map $\text{tr}_{\bar{K}/L}$ creates a commuting diagram.
- Surjectivity of ψ and tr forces $\psi|_L : L \rightarrow L$ to be surjective.

Galois Theoretic Characterization of Real Closed Fields

Proof of Theorem 1.6.1: Steps 3 & 4

(3) $q = 2$ (and $\text{char } K \neq 2$)

- $\text{char } K \neq q \implies \bar{K} = L(a^{1/q})$ for some $a \in L$.
- Let $\beta := a^{1/q^2}$. The norm is $N_{\bar{K}/L}(\beta)^q = (-1)^{q-1}a$.
- Since a is *not* a q -th power in L , **q must be even**.
- Since q is prime, we conclude $q = 2$.

(4) $\bar{K} = L(\sqrt{-1})$

- Let $i = \sqrt{-1} \in \bar{K}$.
- From (3), $\exists a \in L, \beta \in \bar{K}$ with $\beta^4 = a$ and $\bar{K} = L(\beta^2)$.
- Squaring the norm: $N_{\bar{K}/L}(\beta)^2 = -\beta^4 \implies \beta^2 / N_{\bar{K}/L}(\beta) = \sqrt{-1}$.
- $\therefore \sqrt{-1} \notin L$ and $\bar{K} = L(\sqrt{-1})$.

Galois Theoretic Characterization of Real Closed Fields

Proof of Theorem 1.6.1: Steps 5 & 6

(5) $L = K$, thus $\overline{K} = K(i)$

- If $K \neq L$, then $K(i) \neq \overline{K}$.
- Applying steps (1)–(4) to $K(i)$ instead of K leads to a contradiction (it would force another degree-2 extension).
- $\therefore L = K$, which immediately means $\overline{K} = K(i)$.

(6) K is real (and thus real closed)

- Since $i \notin K$, we just need to show the sum of two squares in K is a square (Prop 1.5.2).
- Let $a, b \in K^*$. Since $\overline{K}$ is algebraically closed, $\exists c, d \in K$ with $a + bi = (c + di)^2$.
- Taking conjugates: $a - bi = (c - di)^2$.
- Multiplying them gives: $a^2 + b^2 = (c^2 + d^2)^2$.

$\therefore K$ is a real field. Since $\overline{K} = K(i)$, it has no proper real extensions.